Chapter Contents

Chapter 23 - Steps And Convolution

Chapters 21 and 22 developed forward and inverse Laplace transforms for functions described by one formula. Real inputs often switch on late, switch off again, or change formula at known times. Products of transforms also need a time-domain interpretation.

This chapter introduces two tools for those situations:

- the **unit step function**, which describes switching and delay
- **convolution**, which corresponds to multiplication in the transform domain

We will build piecewise signals, derive the second-shifting theorem, calculate convolutions directly, invert products, and solve convolution-type integral equations. Transforming derivatives and solving differential-equation initial-value problems begins in Chapter 24.

The examples, figures, derivations, and exercises here are independently constructed; the reference chapter is used only as a topic map.

The Unit Step As A Switch

The Goal Of This Block

The goal is to read a shifted unit step as an instruction that changes value at a specified time.

The Definition

The unit step function is:

$$u(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$

A step shifted to time $a > 0$ is:

$$\boxed{u(t-a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a. \end{cases}}.$$

Here a is the switching time. The expression $t - a$ becomes zero when $t = a$.

The exact value assigned at the single point $t = a$ does not affect a Laplace integral. Some books use $u(0) = 1/2$; the transform formulas are unchanged.

Read The Shift Before Calculating

For $u(t-4)$:

- if $t < 4$, then $u(t-4) = 0$
- if $t \geq 4$, then $u(t-4) = 1$

Therefore $u(t-4)$ switches on at $t = 4$.

By contrast, $u(t+4) = u(t-(-4))$ switches at $t = -4$. On the transform domain $t \geq 0$, it is already on.

What to remember:

solve the step's argument equal to zero; that value is the switch location.

Multiplying By A Step

Let $q(t)$ be any function. Then:

$$u(t-a)q(t) = \begin{cases} 0, & t < a, \\ q(t), & t \geq a. \end{cases}$$

The step does not alter the formula after the switch. It suppresses that formula before the switch.

For example:

$$u(t-2)(t^2+1) = \begin{cases} 0, & 0 \leq t < 2, \\ t^2+1, & t \geq 2. \end{cases}$$

A Finite Pulse

Consider:

$$p(t) = u(t-2) - u(t-5).$$

Check the three time intervals separately.

For $0 \leq t < 2$, both steps are off:

$$p(t) = 0 - 0 = 0.$$

For $2 \leq t < 5$, the first step is on and the second is off:

$$p(t) = 1 - 0 = 1.$$

For $t \geq 5$, both steps are on:

$$p(t) = 1 - 1 = 0.$$

Combining the three interval checks gives the pulse formula:

$$\boxed{p(t) = \begin{cases} 0, & 0 \leq t < 2, \\ 1, & 2 \leq t < 5, \\ 0, & t \geq 5. \end{cases}}.$$

The first step switches the pulse on; the second step switches it off.

The Transform Of A Shifted Step

Because $u(t-a) = 0$ before a and 1 after a:

$$\begin{aligned} \mathcal{L}\{u(t-a)\} &= \int_0^\infty e^{-st} u(t-a)\, dt \\ &= \int_a^\infty e^{-st}\, dt. \end{aligned}$$

For $s > 0$, evaluate the improper integral:

$$\begin{aligned} \int_a^\infty e^{-st}\, dt &= \left[-\frac{1}{s} e^{-st} \right]_a^\infty \\ &= 0 - \left(-\frac{1}{s} e^{-as} \right) \\ &= \frac{e^{-as}}{s}. \end{aligned}$$

The evaluated improper integral gives the shifted-step transform:

$$\boxed{\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}, \qquad a > 0, s > 0}.$$

Block 1 Summary

The factor $u(t-a)$ is zero before $t = a$ and one afterward. Differences of steps create finite windows, and a delayed switch contributes the transform factor e^{-as}.

Building Piecewise Signals

The Goal Of This Block

The goal is to convert a piecewise function into one step-function formula and verify that formula on every interval.

The Baseline-And-Changes Method

Start with the formula valid from $t = 0$. At each switching time, add a step multiplied by the required change.

If a signal begins at level L_0, changes by J_1 at $t = a$, and changes by J_2 at $t = b$, write:

$$h(t) = L_0 + J_1 u(t-a) + J_2 u(t-b).$$

The coefficients J_1 and J_2 are **jumps**, not necessarily the new levels.

A Three-Level Example

Represent:

$$h(t) = \begin{cases} 1, & 0 \le t < 2, \\ 4, & 2 \le t < 5, \\ -1, & t \ge 5. \end{cases}$$

The baseline is 1.

At $t = 2$, the level changes from 1 to 4, so the jump is:

$$4 - 1 = 3.$$

At $t = 5$, the level changes from 4 to -1, so the jump is:

$$-1 - 4 = -5.$$

Therefore the step representation is:

$$\boxed{h(t) = 1 + 3u(t-2) - 5u(t-5)}.$$

Verify Every Interval

For $0 \le t < 2$, both steps are zero:

$$h(t) = 1 + 3(0) - 5(0) = 1.$$

For $2 \le t < 5$, only the first step is one:

$$h(t) = 1 + 3(1) - 5(0) = 4.$$

For $t \geq 5$, both steps are one:

$$h(t) = 1 + 3(1) - 5(1) = -1.$$

The compact expression reproduces all three pieces.

A Delayed Ramp That Becomes Flat

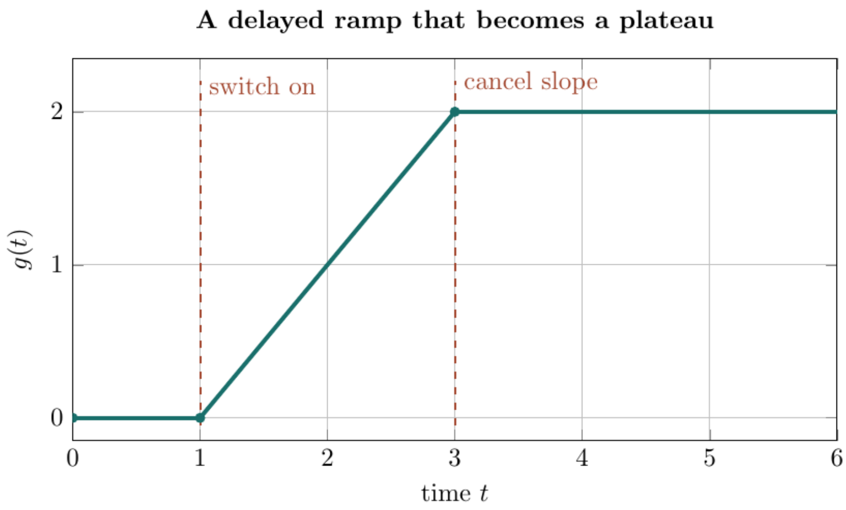

Figure 1: A delayed ramp that becomes a plateau

The graph represents:

$$g(t) = \begin{cases} 0, & 0 \leq t < 1, \\ t - 1, & 1 \leq t < 3, \\ 2, & t \geq 3. \end{cases}$$

Switch on a ramp at $t = 1$:

$$(t-1)u(t-1).$$

At $t = 3$, subtract a new ramp with the same slope:

$$\boxed{g(t) = (t-1)u(t-1) - (t-3)u(t-3)}.$$

The second term cancels the slope after $t = 3$ without cancelling the height already accumulated.

Verify The Ramp And Plateau

For $0 \le t < 1$, both steps are off:

$$g(t) = 0 - 0 = 0.$$

For $1 \le t < 3$, only the first step is on:

$$g(t) = (t-1) - 0 = t - 1.$$

For $t \ge 3$, both steps are on:

$$\begin{aligned} g(t) &= (t-1) - (t-3) \\ &= t - 1 - t + 3 \\ &= 2. \end{aligned}$$

This final cancellation explains the plateau.

Common Construction Errors

Avoid these mistakes:

- using the new level as a step coefficient instead of the change in level
- forgetting that every earlier step remains on at later times
- writing $(t-a)u(t-b)$ when the local formula should start from zero at b
- checking only one interval

The safest test is to substitute step values on every interval.

Block 2 Summary

Build a piecewise signal from its baseline and later changes. For changing slopes, add or subtract shifted local formulas and verify every time interval.

Translation In Time

The Goal Of This Block

The goal is to understand why a delay in time becomes multiplication by e^{-as} in the transform domain.

The Second-Shifting Theorem

Suppose:

$$F(s) = \mathcal{L}\{f(t)\}.$$

Then, for $a > 0$:

$$\boxed{\mathcal{L}\{u(t-a)f(t-a)\} = e^{-as}F(s)}.$$

Reading the same pair backward gives:

$$\boxed{\mathcal{L}^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)}.$$

The step and the shifted argument must appear together.

Derive The Shift Carefully

Because the step is zero before a:

$$\mathcal{L}\{u(t-a)f(t-a)\} = \int_a^\infty e^{-st}f(t-a)\,dt.$$

Use the substitution:

$$v = t-a, \qquad t = v+a, \qquad dt = dv.$$

The lower limit $t = a$ becomes $v = 0$. Substitute into the integral:

$$\begin{aligned}\int_a^\infty e^{-st}f(t-a)\,dt &= \int_0^\infty e^{-s(v+a)}f(v)\,dv \\ &= e^{-as}\int_0^\infty e^{-sv}f(v)\,dv \\ &= e^{-as}F(s).\end{aligned}$$

This derivation shows exactly where the exponential delay factor comes from.

An Inverse Delayed-Sine Example

Find:

$$\mathcal{L}^{-1}\left\{e^{-3s}\frac{2}{s^2+4}\right\}.$$

First identify the unshifted inverse:

$$F(s) = \frac{2}{s^2 + 2^2} \implies f(t) = \sin(2t).$$

The factor e^{-3s} gives a delay of $a = 3$. Replace t by $t - 3$ inside f and attach the step $u(t - 3)$:

$$\boxed{\mathcal{L}^{-1}\left\{e^{-3s}\frac{2}{s^2 + 4}\right\} = u(t - 3)\sin(2(t - 3))}.$$

A Forward Delayed-Polynomial Example

Find the transform of:

$$u(t - 4)(t - 4)^2.$$

Use $f(t) = t^2$, whose transform is:

$$F(s) = \mathcal{L}\{t^2\} = \frac{2}{s^3}.$$

Apply the delay $a = 4$:

$$\boxed{\mathcal{L}\{u(t - 4)(t - 4)^2\} = e^{-4s}\frac{2}{s^3}}.$$

When The Formula Is Not Written Locally

Find $\mathcal{L}\{u(t-2)t^2\}$.

The second-shifting theorem needs a function of $t-2$, but the given factor is t^2. Write:

$$t = (t-2) + 2.$$

Square both sides:

$$\begin{aligned} t^2 &= \left((t-2)+2\right)^2 \\ &= (t-2)^2 + 4(t-2) + 4. \end{aligned}$$

Substituting the rewritten square into the switched expression gives:

$$u(t-2)t^2 = u(t-2)\left[(t-2)^2 + 4(t-2) + 4\right].$$

The corresponding unshifted function is $f(v) = v^2+4v+4$, with transform:

$$F(s) = \frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s}.$$

Apply the delay factor:

$$\boxed{\mathcal{L}\{u(t-2)t^2\} = e^{-2s}\left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s}\right)}.$$

Transform The Ramp-Plateau Signal

Block 2 found:

$$g(t) = (t-1)u(t-1) - (t-3)u(t-3).$$

Each term is a delayed copy of $f(t) = t$, for which $F(s) = 1/s^2$. Apply the second-shifting theorem to each delay:

$$\mathcal{L}\{(t-1)u(t-1)\} = \frac{e^{-s}}{s^2},$$
$$\mathcal{L}\{(t-3)u(t-3)\} = \frac{e^{-3s}}{s^2}.$$

Use linearity:

$$\boxed{\mathcal{L}\{g(t)\} = \frac{e^{-s} - e^{-3s}}{s^2}}.$$

The Most Common Shift Mistake

The theorem applies directly to:

$$u(t-a)f(t-a),$$

not to $u(t-a)f(t)$.

If the expression is $u(t-a)f(t)$, set $v = t-a$, so $t = v+a$. The local function becomes $f(v+a)$:

$$\boxed{\mathcal{L}\{u(t-a)f(t)\} = e^{-as}\mathcal{L}\{f(t+a)\}}.$$

The t inside the transform on the right is a fresh dummy time variable.

Block 3 Summary

A factor e^{-as} represents a delay only when the time-domain result is both switched by $u(t-a)$ and shifted internally from t to $t-a$.

Convolution In The Time Domain

The Goal Of This Block

The goal is to calculate convolution directly and understand how one function is combined with shifted copies of another.

The Definition

For functions f and g defined on $t \geq 0$, their convolution is:

$$\boxed{(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau}.$$

Here:

- t is the output time
- τ is a dummy integration variable
- $g(t-\tau)$ is a shifted and reversed copy of g
- the star $*$ means convolution, not ordinary multiplication

The upper limit is t, so the amount of overlap changes with time.

A First Polynomial Convolution

Let:

$$f(t) = 1, \qquad g(t) = t^2.$$

Insert $f(\tau) = 1$ and $g(t-\tau) = (t-\tau)^2$ into the definition:

$$(f * g)(t) = \int_0^t 1 \cdot (t-\tau)^2 \, d\tau.$$

Expand the square:

$$(t-\tau)^2 = t^2 - 2t\tau + \tau^2.$$

Integrate term by term with respect to τ:

$$\begin{aligned}
(f * g)(t) &= \int_0^t (t^2 - 2t\tau + \tau^2) \, d\tau \\
&= \left[t^2\tau - t\tau^2 + \frac{\tau^3}{3} \right]_0^t \\
&= t^3 - t^3 + \frac{t^3}{3} \\
&= \boxed{\frac{t^3}{3}}.
\end{aligned}$$

Why Convolution Is Commutative

Start from:

$$(f * g)(t) = \int_0^t f(\tau) g(t-\tau) \, d\tau.$$

Use $v = t - \tau$, so $d\tau = -dv$. The limits change as follows:

$$\tau = 0 \Rightarrow v = t, \qquad \tau = t \Rightarrow v = 0.$$

Substitute and reverse the limits:

$$\begin{aligned}(f * g)(t) &= \int_t^0 f(t-v)g(v)(-dv) \\ &= \int_0^t g(v)f(t-v)\,dv \\ &= (g * f)(t).\end{aligned}$$

The change of variable proves the commutative property:

$$\boxed{f * g = g * f}.$$

Commutativity lets us choose whichever order makes the integral easier.

Pulse Overlap Produces A Triangle

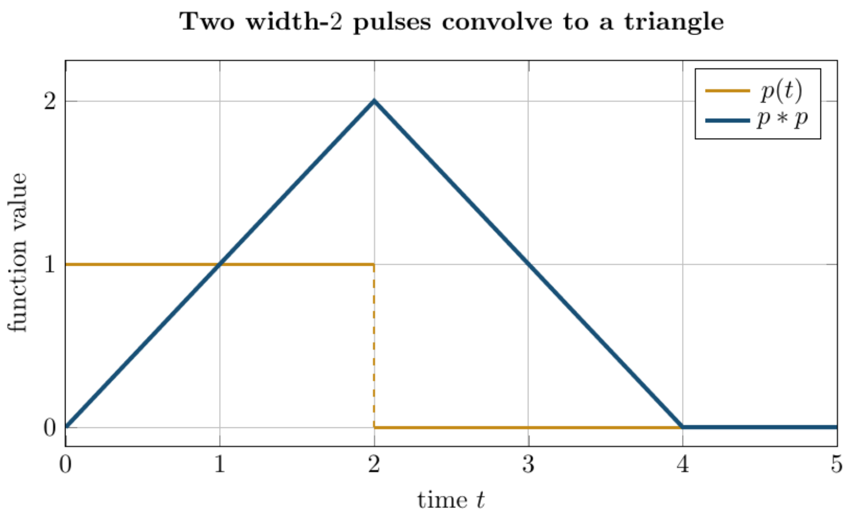

Figure 2: A pulse and the triangular convolution of two identical pulses

Let $p(t) = 1$ on $0 \le t < 2$ and zero otherwise. In $(p * p)(t)$, the integrand equals one wherever two width-2 intervals overlap.

For $0 \le t \le 2$, the overlap length grows from 0 to 2, so:

$$(p * p)(t) = t.$$

For $2 \le t \le 4$, the overlap length shrinks from 2 to 0, so:

$$(p * p)(t) = 4 - t.$$

Outside $0 \le t \le 4$, there is no overlap. Thus:

$$\boxed{(p * p)(t) = \begin{cases} t, & 0 \le t \le 2, \\ 4 - t, & 2 \le t \le 4, \\ 0, & t > 4. \end{cases}}.$$

The key intuition:

convolution measures accumulated overlap, which is why two rectangular pulses produce a triangular output.

An Exponential Convolution

Let $f(t) = e^{-t}$ and $g(t) = e^{-3t}$. Substitute into the definition:

$$(f * g)(t) = \int_0^t e^{-\tau} e^{-3(t-\tau)} \, d\tau.$$

Collect the exponential factors:

$$\begin{aligned} e^{-\tau} e^{-3(t-\tau)} &= e^{-\tau} e^{-3t+3\tau} \\ &= e^{-3t} e^{2\tau}. \end{aligned}$$

Take the factor e^{-3t} outside the τ-integral:

$$\begin{aligned}(f*g)(t) &= e^{-3t}\int_0^t e^{2\tau}\,d\tau \\ &= e^{-3t}\left[\frac{1}{2}e^{2\tau}\right]_0^t \\ &= \frac{1}{2}e^{-3t}(e^{2t}-1) \\ &= \boxed{\frac{1}{2}(e^{-t}-e^{-3t})}.\end{aligned}$$

Linearity Of Convolution

Convolution distributes over sums:

$$f*(g+h)=f*g+f*h.$$

Constants can move outside:

$$(cf)*g=c(f*g)=f*(cg).$$

These properties follow by applying ordinary integral linearity to the convolution definition.

Block 4 Summary

Convolution integrates $f(\tau)g(t-\tau)$ from 0 to t. It is commutative and linear, and it can be interpreted as accumulated overlap.

The Convolution Theorem

The Goal Of This Block

The goal is to connect multiplication in the transform domain with convolution in the time domain.

The Theorem

If:

$$F(s) = \mathcal{L}\{f(t)\}, \qquad G(s) = \mathcal{L}\{g(t)\},$$

then, under the usual convergence assumptions:

$$\boxed{\mathcal{L}\{f * g\} = F(s)G(s)}.$$

Reading the theorem backward:

$$\boxed{\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t)}.$$

This is not the false rule $\mathcal{L}^{-1}\{FG\} = \mathcal{L}^{-1}\{F\}\mathcal{L}^{-1}\{G\}$. The time-domain operation is convolution, not ordinary multiplication.

Why The Product Appears

Start from the transform of the convolution:

$$\mathcal{L}\{f * g\} = \int_0^\infty e^{-st}\left[\int_0^t f(\tau)g(t-\tau)\,d\tau\right]dt.$$

Use $v = t - \tau$, so $t = \tau + v$. The triangular region $0 \le \tau \le t < \infty$ becomes the first quadrant $\tau \ge 0$, $v \ge 0$. Under conditions that permit changing the order:

$$\begin{aligned}
\mathcal{L}\{f * g\} &= \int_0^\infty \int_0^\infty e^{-s(\tau+v)} f(\tau) g(v)\, dv\, d\tau \\
&= \left(\int_0^\infty e^{-s\tau} f(\tau)\, d\tau\right)\left(\int_0^\infty e^{-sv} g(v)\, dv\right) \\
&= F(s)G(s).
\end{aligned}$$

The kernel separates because $e^{-s(\tau+v)} = e^{-s\tau}e^{-sv}$.

An Inverse Product With A Constant Function

Find:

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s+2)}\right\}$$

using convolution.

Factor the transform as:

$$\frac{1}{s(s+2)} = \frac{1}{s}\frac{1}{s+2}.$$

Identify the inverse of each factor:

$$\frac{1}{s} \longleftrightarrow 1, \qquad \frac{1}{s+2} \longleftrightarrow e^{-2t}.$$

Therefore the required inverse is $1 * e^{-2t}$:

$$\begin{aligned}
(1 * e^{-2t})(t) &= \int_0^t 1 \cdot e^{-2(t-\tau)}\, d\tau \\
&= e^{-2t} \int_0^t e^{2\tau}\, d\tau \\
&= e^{-2t} \left[\frac{1}{2} e^{2\tau}\right]_0^t \\
&= \frac{1}{2}(1 - e^{-2t}).
\end{aligned}$$

The convolution calculation gives the inverse product:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{s(s+2)}\right\} = \frac{1}{2}(1-e^{-2t})}.$$

A Product Of Two Trigonometric Transforms

Find:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\}.$$

Write the transform as a product:

$$\frac{1}{(s^2+1)^2} = \frac{1}{s^2+1}\frac{1}{s^2+1}.$$

Since $1/(s^2+1) \longleftrightarrow \sin t$, the inverse is:

$$(\sin * \sin)(t) = \int_0^t \sin\tau \sin(t-\tau)\, d\tau.$$

Use the product-to-sum identity:

$$\sin\tau\sin(t-\tau) = \frac{1}{2}\left[\cos(2\tau - t) - \cos t\right].$$

Integrate both terms:

$$\begin{aligned}
(\sin * \sin)(t) &= \frac{1}{2}\int_0^t [\cos(2\tau-t) - \cos t]\, d\tau \\
&= \frac{1}{2}\left[\frac{1}{2}\sin(2\tau - t) - \tau\cos t\right]_0^t \\
&= \frac{1}{2}\left[\sin t - t\cos t\right].
\end{aligned}$$

The completed trigonometric convolution gives:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\} = \frac{1}{2}(\sin t - t\cos t)}.$$

Combine Convolution With Delay

Find:

$$\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s(s+3)}\right\}.$$

First ignore the delay and invert $1/[s(s+3)]$ by convolution:

$$\begin{aligned} f(t) &= (1 * e^{-3t})(t) \\ &= \int_0^t e^{-3(t-\tau)}\,d\tau \\ &= \frac{1}{3}(1-e^{-3t}). \end{aligned}$$

Now apply the factor e^{-2s}. Replace t by $t-2$ inside f and attach $u(t-2)$:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s(s+3)}\right\} = \frac{1}{3}u(t-2)\left[1-e^{-3(t-2)}\right]}.$$

Convolution handles the product; the second-shifting theorem handles the delay.

Block 5 Summary

Products of transforms correspond to convolutions of inverse transforms. Perform the undelayed inversion first, then apply any exponential delay factor.

Convolution Integral Equations

The Goal Of This Block

The goal is to convert an unknown function inside a convolution integral into an algebraic equation for its transform.

The General Pattern

Consider:

$$y(t) = r(t) + \lambda \int_0^t k(t-\tau)y(\tau)\,d\tau.$$

The integral is $(k * y)(t)$, so:

$$y = r + \lambda(k * y).$$

Let:

$$Y(s) = \mathcal{L}\{y(t)\}, \quad R(s) = \mathcal{L}\{r(t)\}, \quad K(s) = \mathcal{L}\{k(t)\}.$$

Transform both sides and apply the convolution theorem:

$$Y = R + \lambda KY.$$

Move the term containing Y to the left and factor:

$$Y - \lambda KY = R,$$
$$Y(1 - \lambda K) = R.$$

Solving the factored equation for the unknown transform gives:

$$\boxed{Y(s) = \frac{R(s)}{1 - \lambda K(s)}}.$$

An Equation With A Linear Kernel

Solve:

$$y(t) = t + \int_0^t (t - \tau)y(\tau)\,d\tau.$$

The integral is a convolution with $k(t) = t$. Transform the known functions:

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \qquad K(s) = \frac{1}{s^2}.$$

Let $Y(s) = \mathcal{L}\{y(t)\}$. Transform the equation:

$$Y = \frac{1}{s^2} + \frac{1}{s^2}Y.$$

Move Y/s^2 to the left:

$$Y\left(1 - \frac{1}{s^2}\right) = \frac{1}{s^2}.$$

Solve for Y one step at a time:

$$\begin{aligned} Y &= \frac{1/s^2}{1 - 1/s^2} \\ &= \frac{1/s^2}{(s^2-1)/s^2} \\ &= \frac{1}{s^2-1}. \end{aligned}$$

Use $\mathcal{L}\{\sinh t\} = 1/(s^2-1)$:

$$\boxed{y(t) = \sinh t}.$$

Verify The Integral Equation Directly

For $y(t) = \sinh t$, the convolution integral is:

$$I(t) = \int_0^t (t-\tau)\sinh\tau\,d\tau.$$

Split the factor $t - \tau$:

$$I(t) = t\int_0^t \sinh\tau\,d\tau - \int_0^t \tau\sinh\tau\,d\tau.$$

Evaluate the two integrals:

$$\int_0^t \sinh\tau\,d\tau = \cosh t - 1,$$

and, by integration by parts:

$$\int_0^t \tau\sinh\tau\,d\tau = t\cosh t - \sinh t.$$

Substitute them into $I(t)$:

$$\begin{aligned} I(t) &= t(\cosh t - 1) - (t\cosh t - \sinh t) \\ &= \sinh t - t. \end{aligned}$$

The right side of the original equation becomes:

$$t + I(t) = t + (\sinh t - t) = \sinh t = y(t).$$

The solution satisfies the integral equation.

An Equation With An Exponential Kernel

Solve:

$$y(t) = 2 + \int_0^t e^{-(t-\tau)} y(\tau)\, d\tau.$$

Here $r(t) = 2$ and $k(t) = e^{-t}$, so:

$$R(s) = \frac{2}{s}, \qquad K(s) = \frac{1}{s+1}.$$

Transform the equation:

$$Y = \frac{2}{s} + \frac{1}{s+1} Y.$$

Move the convolution term and factor Y:

$$Y\left(1 - \frac{1}{s+1}\right) = \frac{2}{s}.$$

Simplify the factor:

$$1 - \frac{1}{s+1} = \frac{s}{s+1}.$$

Solve for Y:

$$\begin{aligned} Y &= \frac{2/s}{s/(s+1)} \\ &= \frac{2(s+1)}{s^2} \\ &= \frac{2}{s} + \frac{2}{s^2}. \end{aligned}$$

Invert term by term:

$$\boxed{y(t) = 2 + 2t}.$$

Block 6 Summary

Recognise the integral as a convolution, transform it into a product, solve the resulting algebraic equation for $Y(s)$, and verify when practical.

Choosing The Right Route

The Goal Of This Block

The goal is to decide whether a problem calls for interval checking, translation, direct convolution, or the convolution theorem.

A Reliable Workflow

Use this decision sequence:

1. If the function changes at known times, build or inspect unit steps.
2. If e^{-as} appears, invert the undelayed factor before translating.
3. If a time expression contains $u(t-a)f(t)$, rewrite it using $t-a$.
4. If a transform is a useful product $F(s)G(s)$, consider convolution.
5. If an unknown occurs inside $\int_0^t k(t-\tau)y(\tau)d\tau$, transform the entire integral equation and solve for $Y(s)$.
6. Verify step representations interval by interval and transform answers by applying the forward theorem.

Shift And Convolution Must Not Be Confused

The factor e^{-as} signals a time delay:

$$e^{-as}F(s) \longleftrightarrow u(t-a)f(t-a).$$

A product without that exponential signals convolution:

$$F(s)G(s) \longleftrightarrow (f*g)(t).$$

If both occur, convolve first to obtain the undelayed function, then delay the result.

Common Errors

Check for these frequent mistakes:

- treating $u(t-a)$ as one before $t=a$
- using final levels instead of jumps in a step representation
- writing $e^{-as}F(s)$ for $u(t-a)f(t)$ without rewriting the local function
- forgetting the argument $t-\tau$ in a convolution
- multiplying inverse transforms instead of convolving them
- leaving Y on both sides of a transformed integral equation
- applying the time delay to only part of the reconstructed function

Block 7 Summary

Steps control switching, exponential factors control delay, and transform products control convolution. Identify the structure before calculating.

Original Practice Set

Problems 1 To 4: Steps And Translation

Problem 1: Read A Step Combination. Write $h(t) = 2 - 3u(t-1) + 5u(t-4)$ as a piecewise function.

Problem 2: Build A Three-Level Signal. Write with unit steps a function that is 0 for $0 \le t < 2$, 4 for $2 \le t < 5$, and 1 for $t \ge 5$.

Problem 3: Transform A Delayed Sine. Find $\mathcal{L}\{u(t-3)\sin(4(t-3))\}$.

Problem 4: Invert A Delayed Exponential. Find $\mathcal{L}^{-1}\{e^{-2s}/(s+5)\}$.

Problems 5 To 8: Convolution And Products

Problem 5: Rewrite Before Transforming. Find $\mathcal{L}\{u(t-2)t\}$.

Problem 6: Convolve Two Powers. Find $(1 * t)(t)$ directly from the definition.

Problem 7: Convolve Two Exponentials. Find $(e^{-t} * e^{-2t})(t)$.

Problem 8: Invert A Product. Use convolution to find $\mathcal{L}^{-1}\{1/[s(s^2+9)]\}$.

Problems 9 To 12: Combined Methods

Problem 9: Convolve A Repeated Trigonometric Factor. Find $\mathcal{L}^{-1}\{1/(s^2+4)^2\}$ by convolution.

Problem 10: Delay An Inverse Product. Find $\mathcal{L}^{-1}\{e^{-s}/[s(s+2)]\}$.

Problem 11: Solve An Integral Equation. Solve:

$$y(t) = 1 + 2\int_0^t y(\tau)\,d\tau.$$

Problem 12: Diagnose A Translation Error. A learner claims $\mathcal{L}\{u(t-3)f(t)\} = e^{-3s}F(s)$. Explain the error and give the correct general relation.

Worked Answers: Problems 1 To 4

Answer 1. Before $t = 1$, both steps are zero, so $h = 2$. Between 1 and 4, only the first step is one, so $h = 2 - 3 = -1$. After $t = 4$, both are one, so $h = 2 - 3 + 5 = 4$. Therefore:

$$\boxed{h(t) = \begin{cases} 2, & 0 \le t < 1, \\ -1, & 1 \le t < 4, \\ 4, & t \ge 4. \end{cases}}$$

Answer 2. The baseline is 0. The jump at 2 is $4 - 0 = 4$, and the jump at 5 is $1 - 4 = -3$. Hence:

$$\boxed{h(t) = 4u(t-2) - 3u(t-5)}.$$

Answer 3. Use $\mathcal{L}\{\sin(4t)\} = 4/(s^2 + 16)$ and delay by 3:

$$\boxed{\mathcal{L}\{u(t-3)\sin(4(t-3))\} = e^{-3s}\frac{4}{s^2+16}}$$

Answer 4. The undelayed inverse is $\mathcal{L}^{-1}\{1/(s+5)\} = e^{-5t}$. Apply the delay $a = 2$ to the entire function:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s+5}\right\} = u(t-2)e^{-5(t-2)}}.$$

Worked Answers: Problems 5 To 8

Answer 5. Rewrite $t = (t-2) + 2$:

$$u(t-2)t = u(t-2)[(t-2)+2].$$

The unshifted transform is $1/s^2 + 2/s$, so:

$$\boxed{\mathcal{L}\{u(t-2)t\} = e^{-2s}\left(\frac{1}{s^2} + \frac{2}{s}\right)}.$$

Answer 6. Use $f(\tau) = 1$ and $g(t-\tau) = t - \tau$:

$$\begin{aligned}(1 * t)(t) &= \int_0^t (t-\tau)\,d\tau \\ &= \left[t\tau - \frac{\tau^2}{2}\right]_0^t \\ &= \boxed{\frac{t^2}{2}}.\end{aligned}$$

Answer 7. Substitute the two exponentials into the convolution definition:

$$\begin{aligned}(e^{-t} * e^{-2t})(t) &= \int_0^t e^{-\tau}e^{-2(t-\tau)}\,d\tau \\ &= e^{-2t}\int_0^t e^{\tau}\,d\tau \\ &= e^{-2t}(e^t - 1) \\ &= \boxed{e^{-t} - e^{-2t}}.\end{aligned}$$

Answer 8. Use $1/s \longleftrightarrow 1$ and $1/(s^2+9) \longleftrightarrow (1/3)\sin(3t)$:

$$\begin{aligned} f(t) &= \frac{1}{3}\int_0^t \sin(3(t-\tau))\,d\tau \\ &= \frac{1}{3}\left[\frac{1}{3}\cos(3(t-\tau))\right]_0^t \\ &= \boxed{\frac{1-\cos(3t)}{9}}. \end{aligned}$$

Worked Answers: Problems 9 To 12

Answer 9. Since $1/(s^2+4) \longleftrightarrow (1/2)\sin(2t)$:

$$f(t) = \frac{1}{4}\int_0^t \sin(2\tau)\sin(2(t-\tau))\,d\tau.$$

Using product-to-sum and integrating gives:

$$\begin{aligned} f(t) &= \frac{1}{8}\int_0^t [\cos(4\tau-2t)-\cos(2t)]\,d\tau \\ &= \frac{1}{8}\left[\frac{1}{4}\sin(4\tau-2t)-\tau\cos(2t)\right]_0^t \\ &= \boxed{\frac{\sin(2t)-2t\cos(2t)}{16}}. \end{aligned}$$

Answer 10. First invert $1/[s(s+2)]$ by convolution:

$$f(t) = \frac{1}{2}(1-e^{-2t}).$$

Then apply the delay $a = 1$:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s(s+2)}\right\} = \frac{1}{2}u(t-1)[1-e^{-2(t-1)}]}.$$

Answer 11. Write the integral as $(1 * y)(t)$. Transform the equation:

$$Y = \frac{1}{s} + \frac{2}{s}Y.$$

Collect the Y terms and solve:

$$\begin{aligned} Y\left(1 - \frac{2}{s}\right) &= \frac{1}{s}, \\ Y\frac{s-2}{s} &= \frac{1}{s}, \\ Y &= \frac{1}{s-2}. \end{aligned}$$

Inverting $Y(s) = 1/(s-2)$ gives:

$$\boxed{y(t) = e^{2t}}.$$

Answer 12. The second-shifting theorem requires $u(t-3)f(t-3)$, not $u(t-3)f(t)$. Set $v = t-3$, so $t = v+3$:

$$\begin{aligned} \mathcal{L}\{u(t-3)f(t)\} &= \int_3^\infty e^{-st}f(t)\,dt \\ &= e^{-3s}\int_0^\infty e^{-sv}f(v+3)\,dv. \end{aligned}$$

Thus the correct general relation is:

$$\boxed{\mathcal{L}\{u(t-3)f(t)\} = e^{-3s}\mathcal{L}\{f(t+3)\}}.$$

Chapter 23 Final Summary

The central transform pairs are:

$$\begin{aligned} u(t-a)f(t-a) &\longleftrightarrow e^{-as}F(s), \\ (f*g)(t) &\longleftrightarrow F(s)G(s). \end{aligned}$$

For step functions, verify each interval. For translation, shift the internal time argument as well as attaching the step. For convolution, use:

$$(f*g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau.$$

For a convolution integral equation, transform the convolution into a product, isolate $Y(s)$ completely, and then invert.

switches describe when a formula is active, delays move a complete response in time, and convolution describes how two time-domain responses accumulate.